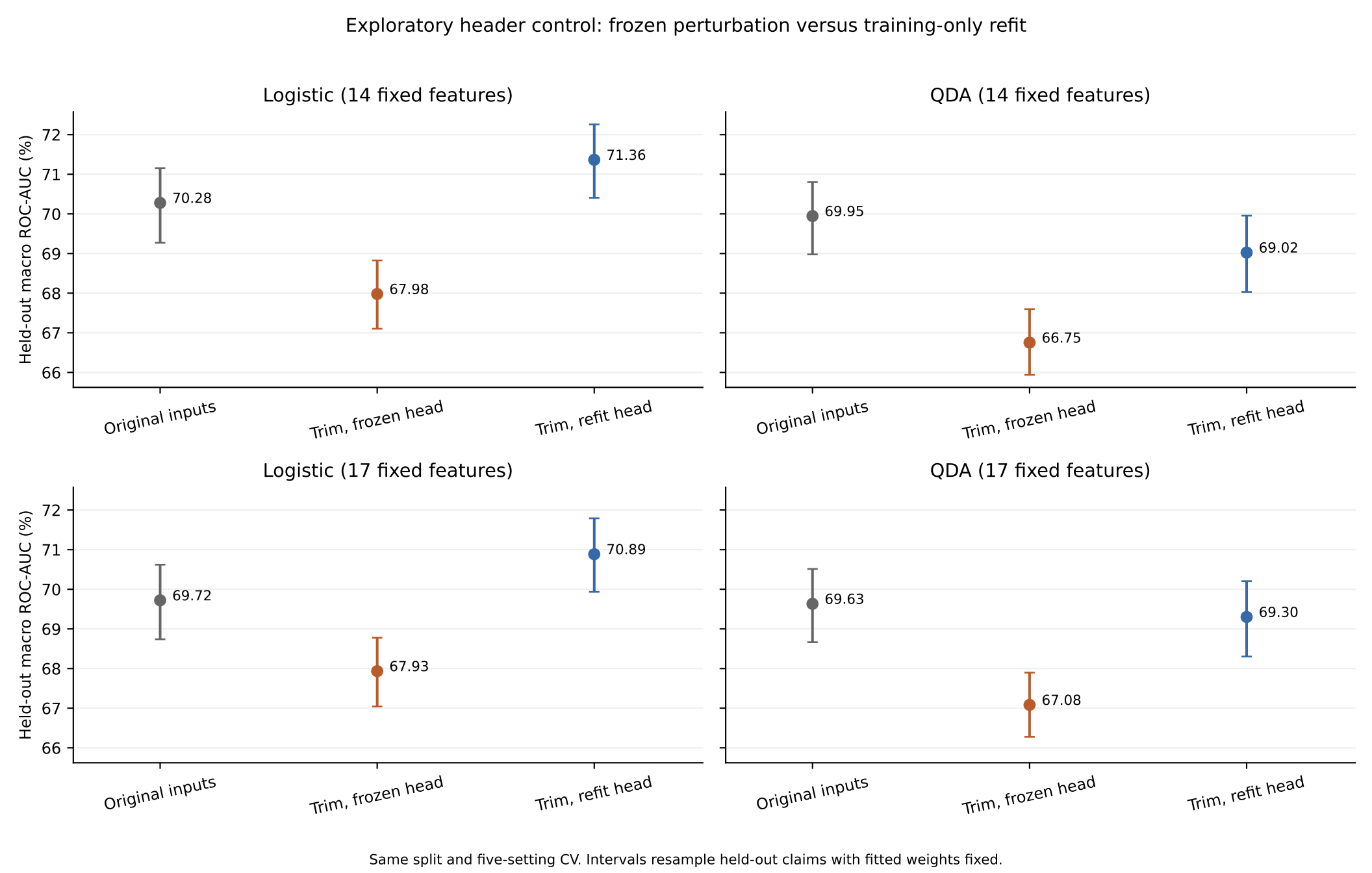
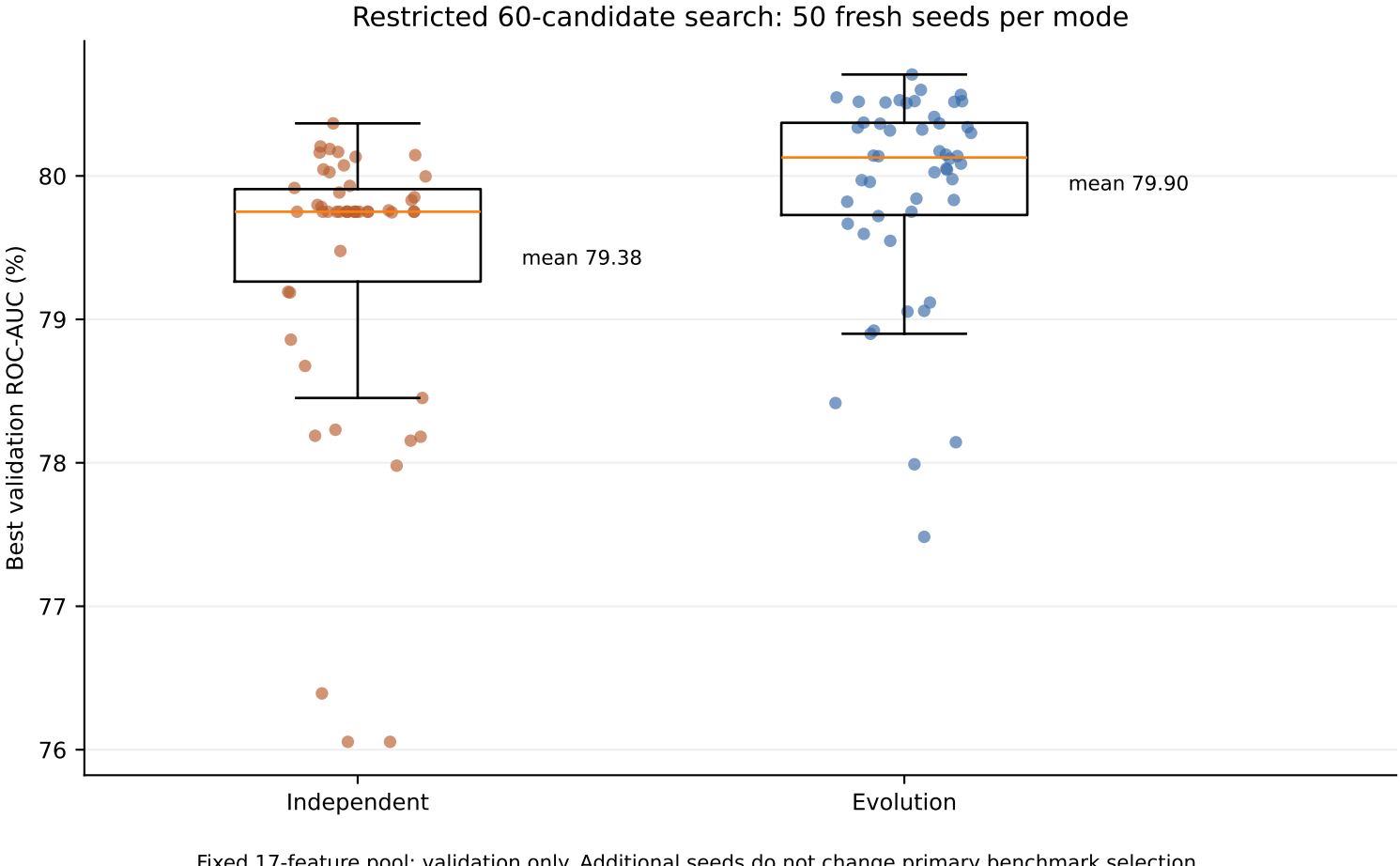
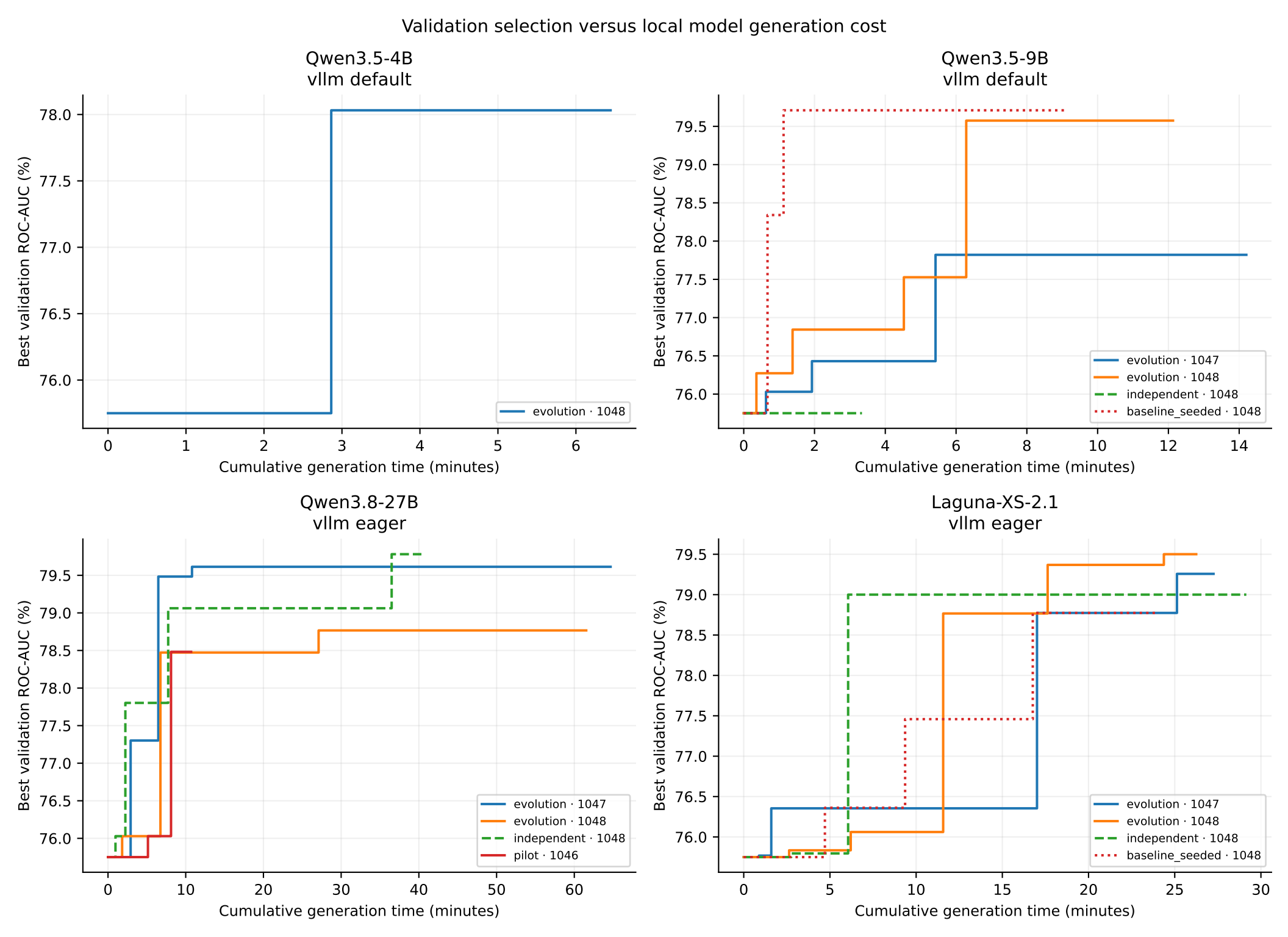
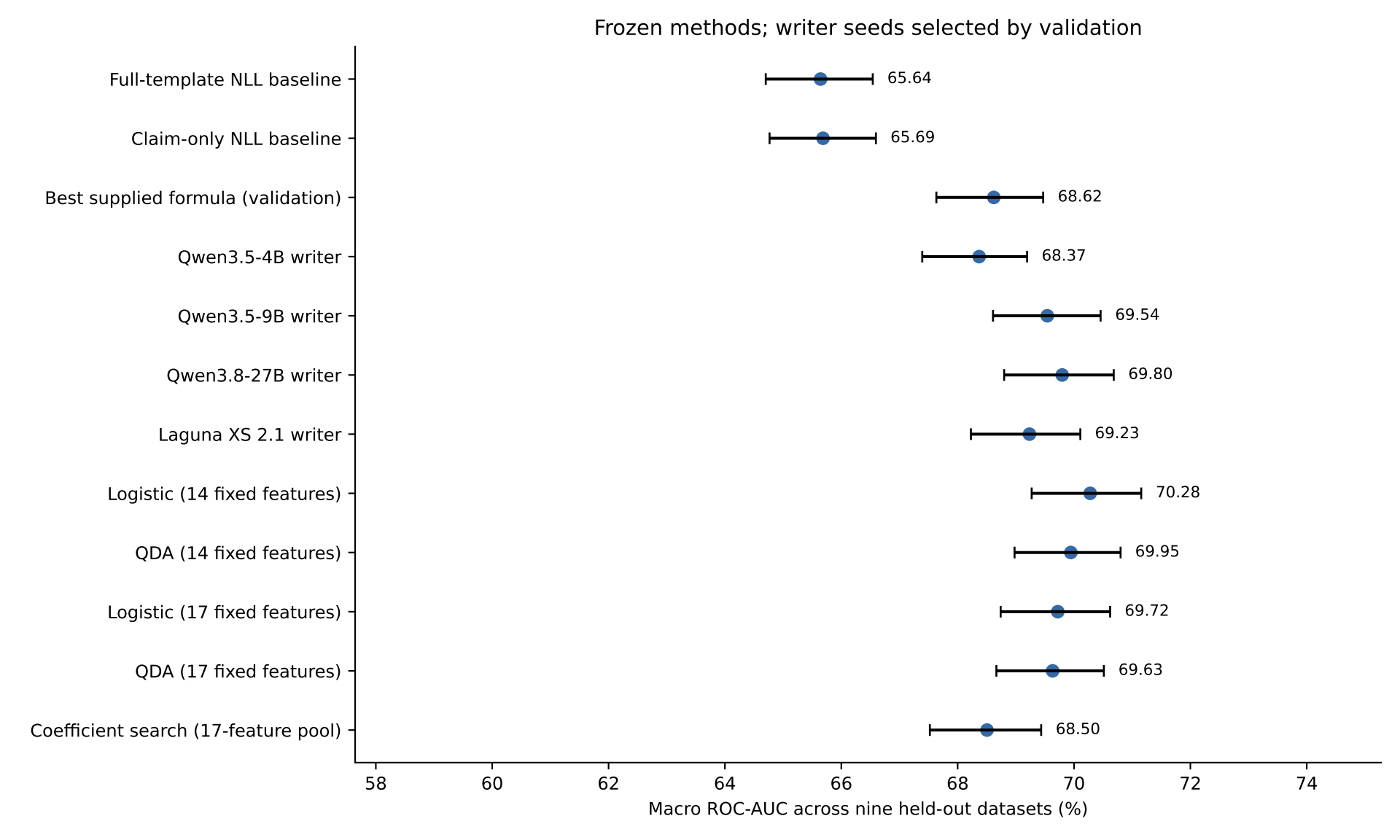
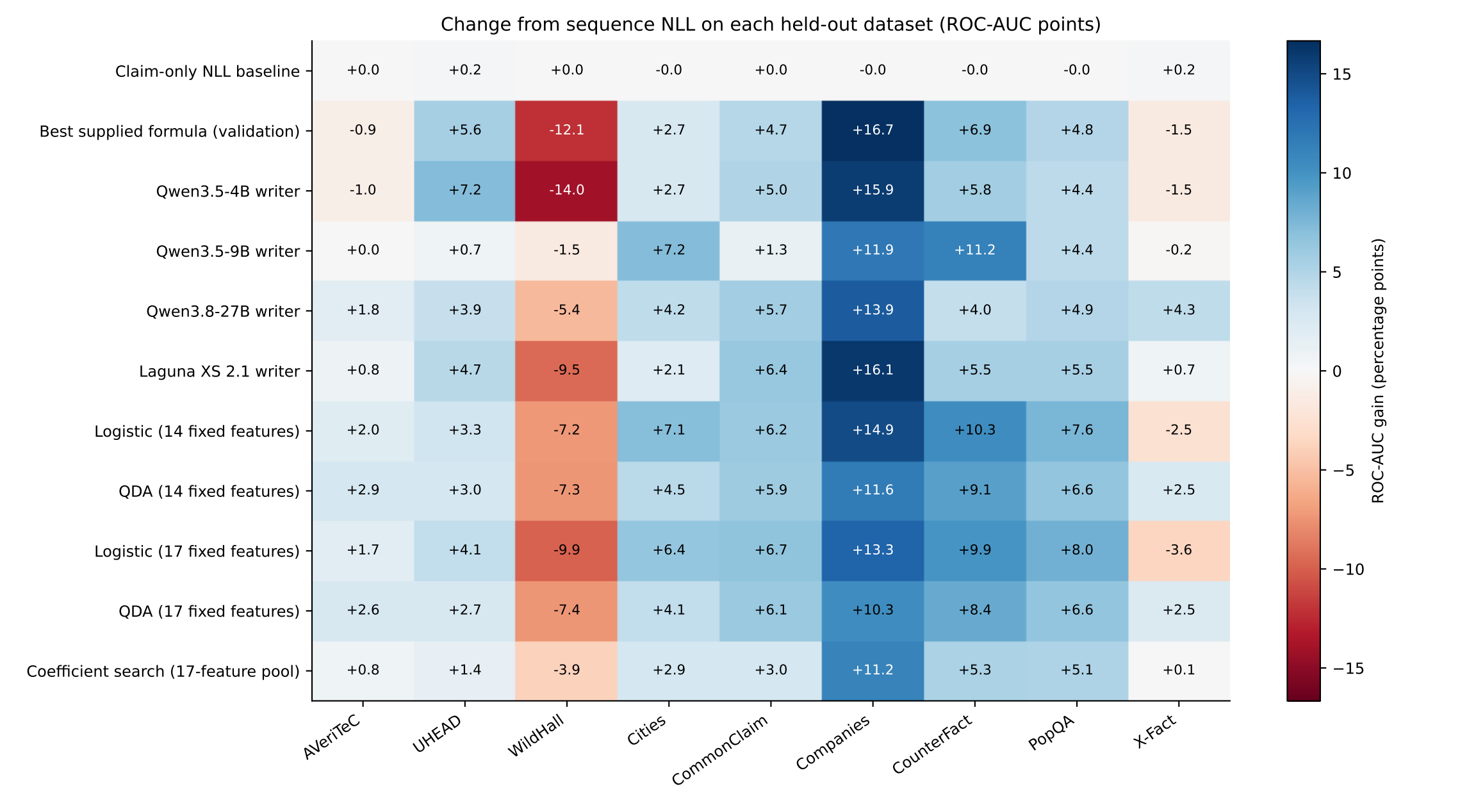
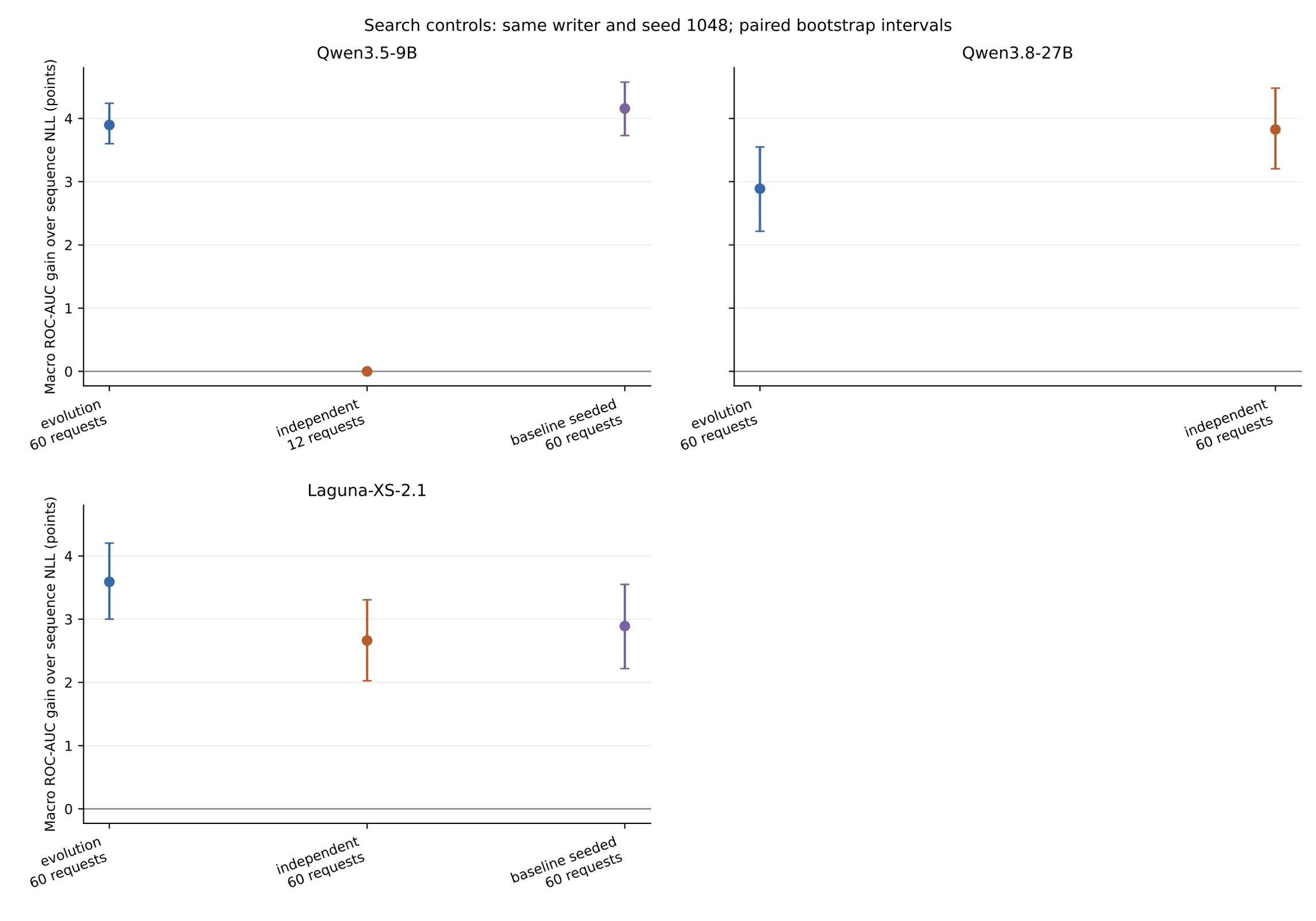

Exploratory header control: frozen perturbation versus training-only refit
Logistic (14 fixed features)
QDA (14 fixed features)
Logistic (17 fixed features)
QDA (17 fixed features)
Held-out macro ROC-AUC (%)
72
71
70
69
68
67
66
70.28
67.98
71.36
69.95
66.75
69.02
69.72
67.93
70.89
69.63
67.08
69.30
Original inputs
Trim, frozen head
Trim, refit head
Same split and five-setting CV. Intervals resample held-out claims with fitted weights fixed.

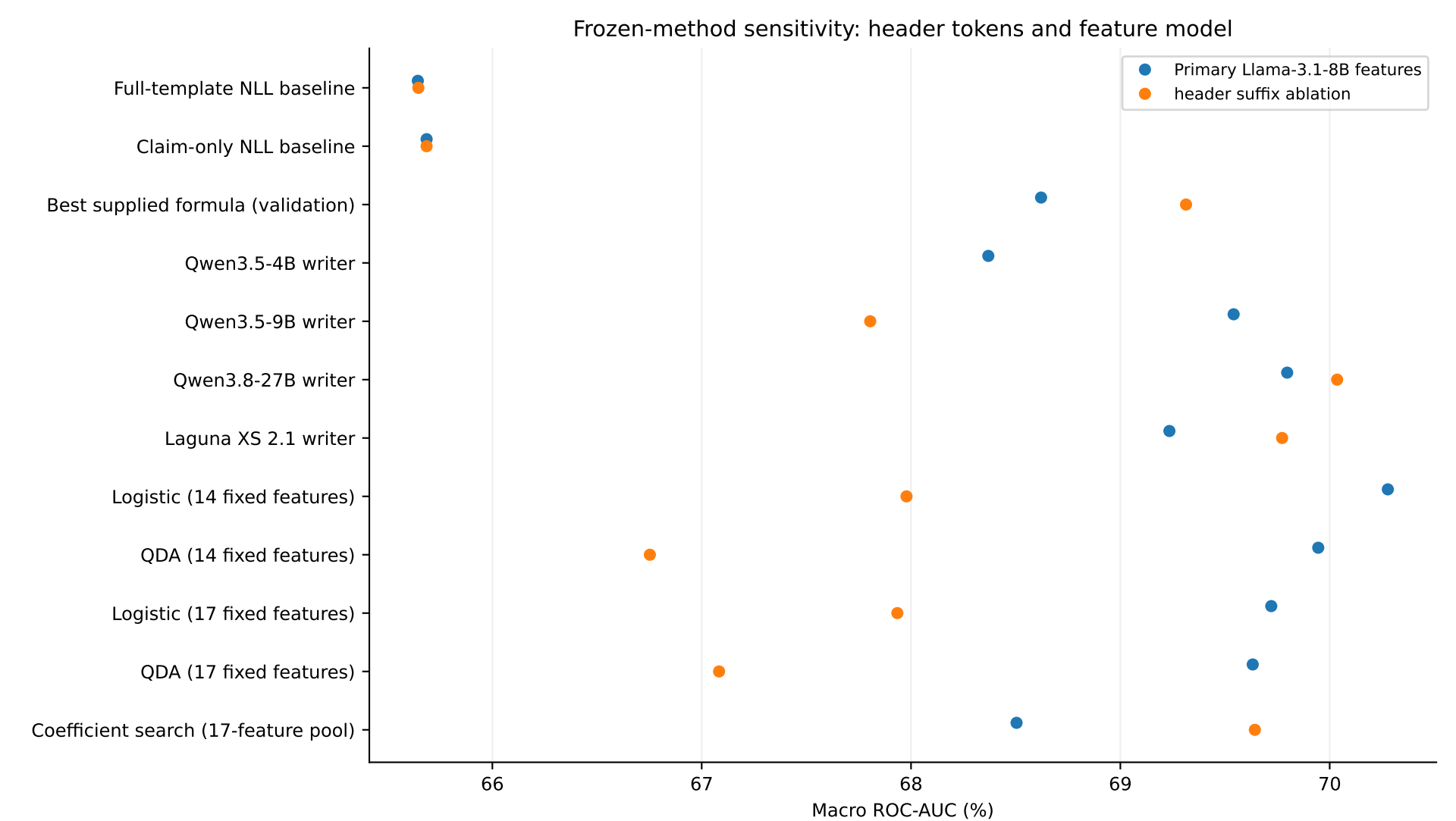

Restricted 60-candidate search: 50 fresh seeds per mode
Best validation ROC-AUC (%)
80
79
78
77
76
mean 79.38
mean 79.90
Independent
Evolution
Fixed 17-feature pool; validation only. Additional seeds do not change primary benchmark selection.


Validation selection versus local model generation cost
Qwen3.5-4B
vllm default
Qwen3.5-9B
vllm default
Qwen3.8-27B
vllm eager
Laguna-XS-2.1
vllm eager
Best validation ROC-AUC (%)
Cumulative generation time (minutes)
evolution · 1048
evolution · 1047
independent · 1048
baseline_seeded · 1048
pilot · 1046

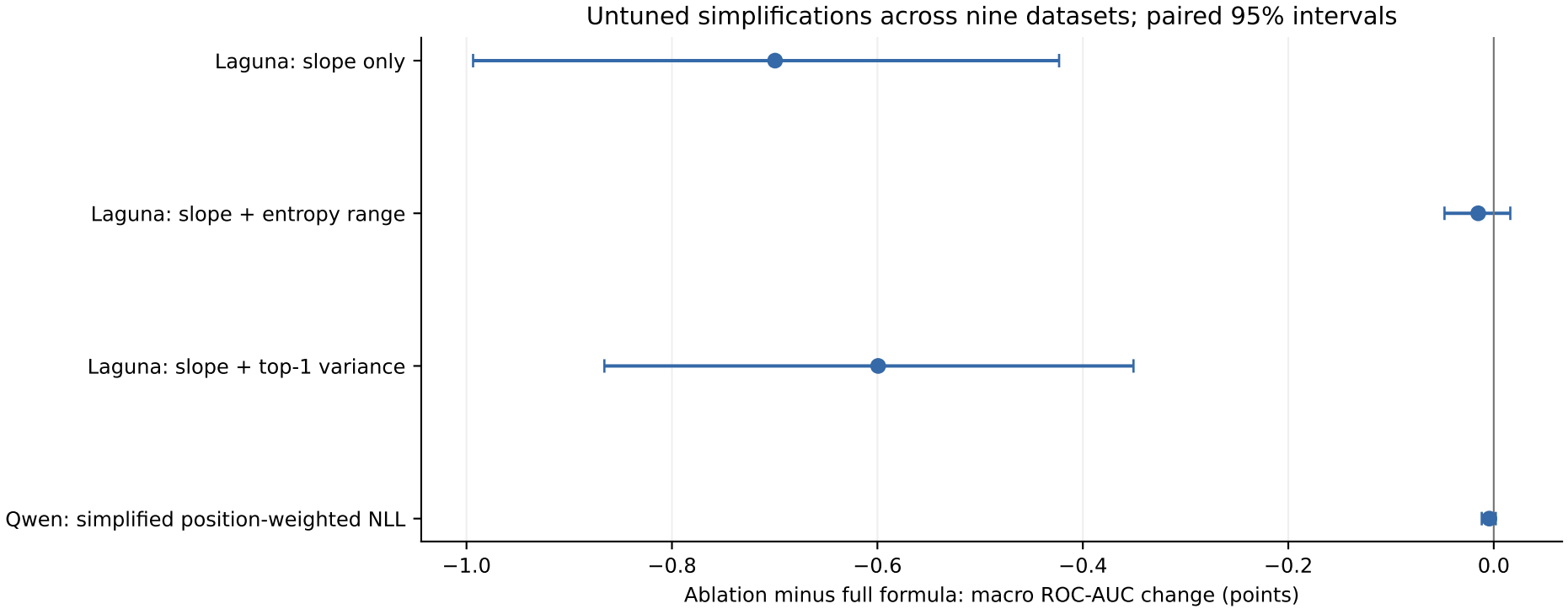

Frozen methods; writer seeds selected by validation
Full-template NLL baseline
65.64
Claim-only NLL baseline
65.69
Best supplied formula (validation)
68.62
Qwen3.5-4B writer
68.37
Qwen3.5-9B writer
69.54
Qwen3.8-27B writer
69.80
Laguna XS 2.1 writer
69.23
Logistic (14 fixed features)
70.28
QDA (14 fixed features)
69.95
Logistic (17 fixed features)
69.72
QDA (17 fixed features)
69.63
Coefficient search (17-feature pool)
68.50
58
60
62
64
66
68
70
72
74
Macro ROC-AUC across nine held-out datasets (%)


Change from sequence NLL on each held-out dataset (ROC-AUC points)
Claim-only NLL baseline
Best supplied formula (validation)
Qwen3.5-4B writer
Qwen3.5-9B writer
Qwen3.8-27B writer
Laguna XS 2.1 writer
Logistic (14 fixed features)
QDA (14 fixed features)
Logistic (17 fixed features)
QDA (17 fixed features)
Coefficient search (17-feature pool)
+0.0 +0.2 +0.0 -0.0 +0.0 -0.0 -0.0 -0.0 +0.2
-0.9 +5.6 -12.1 +2.7 +4.7 +16.7 +6.9 +4.8 -1.5
-1.0 +7.2 -14.0 +2.7 +5.0 +15.9 +5.8 +4.4 -1.5
+0.0 +0.7 -1.5 +7.2 +1.3 +11.9 +11.2 +4.4 -0.2
+1.8 +3.9 -5.4 +4.2 +5.7 +13.9 +4.0 +4.9 +4.3
+0.8 +4.7 -9.5 +2.1 +6.4 +16.1 +5.5 +5.5 +0.7
+2.0 +3.3 -7.2 +7.1 +6.2 +14.9 +10.3 +7.6 -2.5
+2.9 +3.0 -7.3 +4.5 +5.9 +11.6 +9.1 +6.6 +2.5
+1.7 +4.1 -9.9 +6.4 +6.7 +13.3 +9.9 +8.0 -3.6
+2.6 +2.7 -7.4 +4.1 +6.1 +10.3 +8.4 +6.6 +2.5
+0.8 +1.4 -3.9 +2.9 +3.0 +11.2 +5.3 +5.1 +0.1
AVeriTeC
UHEAD
WildHall
Cities
CommonClaim
Companies
CounterFact
PopQA
X-Fact
15
10
5
0
−5
−10
−15
ROC-AUC gain (percentage points)


Search controls: same writer and seed 1048; paired bootstrap intervals
Qwen3.5-9B
Qwen3.8-27B
Laguna-XS-2.1
Macro ROC-AUC gain over sequence NLL (points)
0
1
2
3
4
evolution
60 requests
independent
12 requests
baseline seeded
60 requests
independent
60 requests



Missing markers indicate an unavailable result or a method outside that frozen cohort.


Untuned simplifications across nine datasets; paired 95% intervals
Laguna: slope only
Laguna: slope + entropy range
Laguna: slope + top-1 variance
Qwen: simplified position-weighted NLL
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
Ablation minus full formula: macro ROC-AUC change (points)